

Renewables, Storage, and Energy Affordability: How might the strategic deployment of storage create a more affordable and sustainable grid?

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1 Introduction and Motivation

National electricity prices have increased at double the inflation rate in the United States. Based on data reported by the Energy Information Administration, over half of states are experiencing electricity prices that exceed the Consumer Price Index (see Appendix Figure A1). This has made energy affordability a central concern both for ratepayers and policy makers. One narrative that has recently traction is that the increase of renewable generation and “green policy” are principally responsible for the increases in residential electricity prices.¹ As a result, incentive programs such as the Inflation Reduction Act and the Bipartisan Infrastructure Law have both been rolled-back by the current federal administration. These decisions will have substantial impacts on the future cost of energy.

The true impacts of renewables on prices are unclear. They provide several benefits to the grid such as low marginal operating costs, congestion alleviation if strategically located, and the capital costs associated with developing them are relatively low as compared to a fossil fuel plant (International Renewable Energy Agency (IRENA), 2025). They also create opportunities for distributed generation through rooftop solar and storage systems. However, there also exist several challenges.

The incumbent technologies are centralized, dispatchable fossil fuel plants. Because of this, the development of the grid was arranged to accommodate the transmission of electricity across long distances, and the uni-directional distribution of power towards end users. The rapid proliferation of renewables has created three primary challenges for utilities. First, renewable generation is intermittent which introduces voltage instability and supply uncertainty for grid planners. This may incur cost increases as a result. Next, the integration of renewables requires a substantial build out of transmission and distribution infrastructure, especially for wind energy which is often sited farther away from demand pockets. Third, bidirectional power flows on the distribution network from rooftop solar systems has introduced challenges to the electricity network. (Argosino and Knittel, 2025)

One technology that has shown to be particularly useful for mitigating the challenges associated with renewables is utility-scale storage. Across the state of Texas, storage has been one of the fastest growing generation and grid support resources. Because of its dispatchability, storage can provide both stability and supply to the grid to offset the negative impacts of renewable generation. It may also be used to offset the necessity of building transmission and distribution lines by collocating these resources with high-load buses (Garcia and Smith, 2020).

Taking all these into account, our project aims to address the following questions:

¹Goreham, “Electricity Prices Going Up? Green Policy Is to Blame,” 2025.

- Is the increased generation from renewable technologies responsible for an increase of residential electricity prices?
- What are the potential mechanisms through which increased renewable generation may be impacting grid costs?
- How might utility-scale storage be strategically deployed to mitigate the potential challenges introduced by renewables to the grid?

We break our inquiry up into three sections to address each question. The first is an econometric analysis that aims to understand the impact of changes in generation mixes on residential electricity prices and utility costs. We source data and build upon an empirical strategy from Argosino and Knittel (2025), but our strategy differs as it does not examine the impacts on residential prices as a result of Renewable Portfolio Standards. Rather, we focus exclusively on how prices and utility costs are impacted by changing the generation mix. To validate these results, we build an optimization model using the Texas 2,000 bus synthetic grid data set developed by Lee et al. (2022) and simulate an increase in the amount of renewable generation in the grid. This enables us to understand the grid-scale challenges associated with renewable generation in the region. We then test two strategies for battery collocation that enables us to understand how batteries may alleviate the challenges with increased utility-scale solar generation.

2 Econometric Formulation and Results

An econometric analysis enables us to understand national trends associated with increased renewable generation and its relationship with residential electricity prices. We source data on state-year residential electricity prices and generation mixes from the Energy Information Administration's Form 861 and 923. We also source information on utility capital expenditures (CapEx) and Operations & Maintenance (O&M) costs from the Federal Energy Regulatory Commission's Form 1. A table that contains each of the variables, their sources, and units is included in Appendix Table A1.

We use a fixed effects regression to understand the contributions of different energy generation technologies towards explaining variations in residential electricity prices. The equation takes the following form:

$$\text{Rate}_{st} = \beta_1 \text{Renewables}_{st} + \gamma_s + \delta_t + \mathbf{X}'_{st} \boldsymbol{\theta} + \varepsilon_{st}, \quad (1)$$

where Renewables includes the state-year generation of utility-scale solar, wind, and hydro. These are the categories in which we are primarily interested. γ_s and δ_t include state and year fixed effects, respectively, to control for both interstate and interannual rate heterogeneity that is unobserved in the current data set. These absorb state and year shocks to the energy supply. $\mathbf{X}'_{st} \boldsymbol{\theta}$ includes additional control variables such as other large scale generation technologies. This regression enables us to observe the impact of different generation technologies on the residential electricity prices as a function of changes in the state level generation mix over time. By including state and year fixed effects, we are able to observe within state variation over time to produce plausibly causal estimates.

Figure 1 displays the key results of our regressions. We find that increased utility-scale renewable generation is correlated with lower residential electricity prices. In particular, utility-scale solar and wind have significant negative correlations. This contradicts the narrative that renewables are responsible for greater residential electricity prices. In fact, these results imply that renewable generation may actually cause lower residential electricity prices.

These results are certainly not conclusive, but they warrant further investigation into why renewables may be correlated with lower residential electricity prices. Some potential relevant channels through which

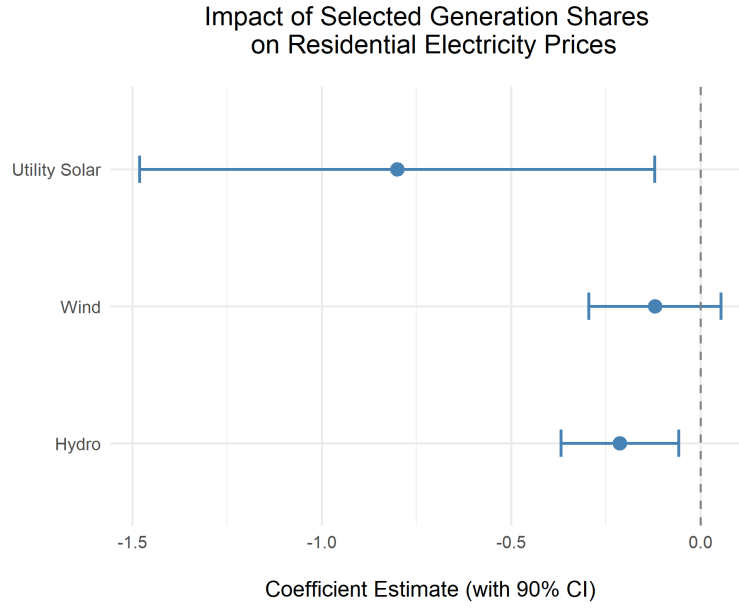


Figure 1: Coefficient plot that corresponds to the fourth column of Appendix Table A2 with state and year fixed effects. Utility-scale solar and hydro are significantly correlated with lower residential electricity prices.

renewables may lower prices is by lowering utility operations and maintenance (O&M) costs. We use data extracted from the FERC's Form 1 survey on O&M cost information on generation, transmission, and distribution as self-reported by utilities. We use a similar regression strategy as before such that our equation takes the following form:

$$\text{Cost}_{st} = \beta_1 \text{Renewables}_{st} + \alpha_u + \gamma_s + \delta_t + \mathbf{X}'_{st}\boldsymbol{\theta} + \varepsilon_{st}, \quad (2)$$

where we add utility fixed effects as α_u to Equation 2 and we substitute the outcome variable with utility generation, transmission, distribution, or total costs per-kWh of electricity sold to ultimate consumers or for resale.² These results are available in Appendix Tables A3, A4, A5, and A6. These results begin to point us towards the possible cost reductions associated with utility-scale renewables. They all correlate with lower generation costs, which is consistent with their low marginal costs of production. Hydro and wind, however, contribute to greater transmission costs, likely because they require longer transmission lines and therefore incur greater O&M costs. Solar is the opposite as these technologies can be located more closely to high load areas than wind, hydro, and large fossil fuel plants. Solar and wind seem to be correlated with lower distribution costs as their peak production hours may coincide with peak load hours, thereby reducing strain from other transmission lines to meet high level load, especially in warm states with high air-conditioning penetration as seen in Figure 2 (Fares and King, 2017). Overall, all utility-scale renewables seem to be correlated with lower utility costs which may be passed onto consumers in the form of lower residential electricity prices. Utility-scale solar, in particular, seems to place the strongest, most statistically significant downward pressure on costs, and will serve as the focus of subsequent analysis.

²We define total as the sum of the three aforementioned categories

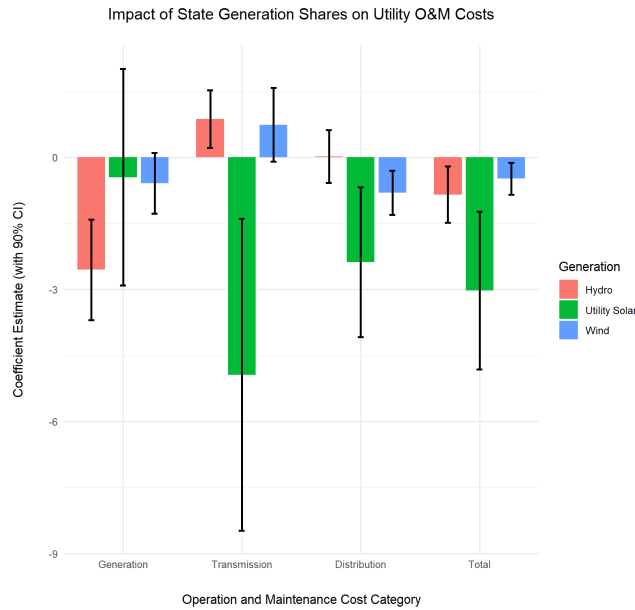


Figure 2: This graph plots the regression results from the following Appendix Tables where the outcome variable is switched to distribution in Table A3, transmission in Table A4, generation in Table A5, and the total in Table A6. Each bar cluster corresponds to a separate regression model and enables us to observe the correlation and significance of utility-scale renewables on different utility cost categories. In general, solar is strongly correlated with lower transmission, distribution, and total costs which suggests that it may also place downward pressure on retail electricity prices.

3 Optimization Modeling of Renewable Penetration

The econometric results provide us with a robust correlation between utility-scale solar and lower residential electricity prices and lower utility costs. This suggests that there may exist a plausibly causal relationship between the increased proliferation of solar and lowering prices. To justify a causal interpretation of these estimates, however, more work is needed to understand exactly how utility-scale solar may reduce utility costs and therefore prices.

An energy systems optimization model is well-suited to address this question because it enables us to use a physics-based approach to understand how changes in generation mix may impact grid-level costs, and locational marginal prices. We use data from Lee et al. (2022) and the MatPower energy systems optimization software with a Gurobi solver in MATLAB (Zimmerman et al., 2011). Lee et al includes subhourly data for the year 2020 in Texas to simulate a 2000-bus synthetic grid model for the Texas transmission network (Birchfield et al., 2017). These data include load profiles for each bus as well as generation profiles for solar, wind, and hydro. We run the included security-constrained unit-commitment (SCUC) script first and then use those results to run a separate security-constrained economic dispatch (SCED) script. As these are standard problem formulations in energy systems optimization and given the brevity of this report, we include a reproduction of the optimization formulation from Lee et al. (2022) for both SCUC and SCED in Appendix B. For this analysis, we focus on the three highest peak load days for the data we have available. The Table 1 displays the summary statistics of each day we modeled for this project.

Table 1: Peak load and total system cost on selected days in 2020

Date	Day Number	Peak Load [MW]	Total System Cost [\$]
July 12, 2020	194	71,743	31,996,476
August 3, 2020	216	74,754	37,096,241
August 12, 2020	225	76,375	37,797,812

The body of this paper will discuss results associated with August 12th (Day 225) since that is the most extreme case in the data set. We dedicate sections in the Appendix to analyzing results from the other two days. We begin our analysis by analyzing grid conditions in the data as it is provided by Lee et al. This provides us with a baseline with which we can compare subsequent models. We then simulate our first original case by inducing a 25% in the solar generation. We develop this case by multiplying the original data’s solar generation by 1.25. If this causes the generator to exceed its specified capacity, we instead set the generation output to its maximum output. For example, if a solar generator was producing 10 MW in some time interval and its maximum production is 12 MW, then a 1.25 increase would result in a production of 12.5 MW. Since this exceeds the maximum production, we instead force this value to its maximum production value of 12 MW. The following sections describe our analysis process and key results.

3.1 Data Extraction and Analysis

To analyze the results of the SCED, we developed a custom data extraction pipeline. The raw simulation results are stored in MatPower’s internal `mdo` (MOST Data Object) structure. We utilize a custom MATLAB script, to parse this structure and export key operational metrics into CSV format for downstream analysis in Python.

For each simulation hour, we extract:

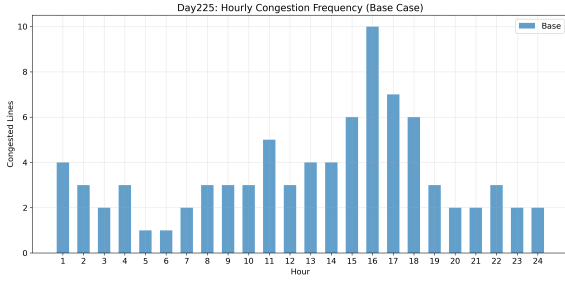
- Locational Marginal Prices (LMP): The shadow price at each bus, representing the marginal cost of serving the next increment of load.
- Generator Dispatch: The real power output (P_g), maximum capacity (P_{max}), and commitment status of every generator.
- Branch Flows and Congestion: The power flow on every transmission line and the associated shadow prices (μ) for binding thermal constraints.

We specifically track *Congestion Frequency*, defined as the total count of branch flow constraints that bind (where the shadow price $\mu > 10^{-9}$) over the simulation period. This allows us to quantify the stress placed on the transmission network by different generation and storage profiles.

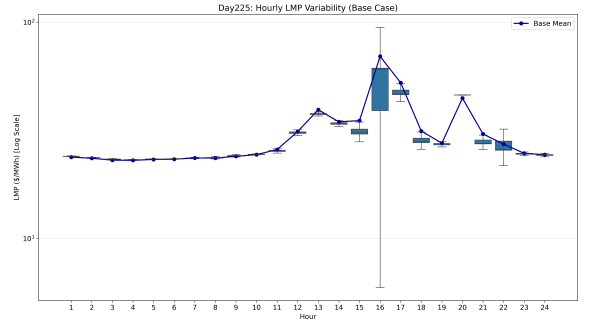
3.2 Results: Base Case Analysis

We first establish a baseline using the peak load day, which was the summer peak for the Texas grid in 2020. In the Base Case, the system operates with the standard generation mix defined in the synthetic grid dataset. The simulation yielded a Total System Cost of \$37,797,812.11 and a system-wide average LMP of \$30.56/MWh. The grid experienced 84 distinct congestion events throughout the day.

Figure 3 illustrates the hourly congestion profile and LMP variability for the Base Case. The base case congestion levels and LMPs serve as the benchmark for comparison with the other cases we create below.



(a) Hourly Congestion Events



(b) LMP Variability overlaid with average LMPs

Figure 3: The Base Case operational metrics showing moderate congestion and price stability.

3.3 Results: High PV Generation

In general, increasing utility-scale solar (High PV) penetration tends to reduce total system costs and Locational Marginal Prices (LMPs), as seen in our simulations for Day 194 and Day 216 (see Appendix C). These days represent typical operating conditions in which the grid can effectively integrate higher renewable generation. On Day 225, however, total system costs increase by \$11,897. Although this constitutes only a 0.03% increase in cost, it is counterintuitive because solar would normally be expected to lower overall costs. This result underscores how unexpected power-flow patterns can emerge when the grid suddenly faces large increases in supply, even when the marginal cost of that production is near zero.

Figure 4 shows that LMP variability decreases in the High PV scenario, as indicated by the narrower interquartile range and shorter whiskers in the box plot. Despite this increased price stability and the influx of zero-marginal-cost energy, total system cost rises on Day 225.

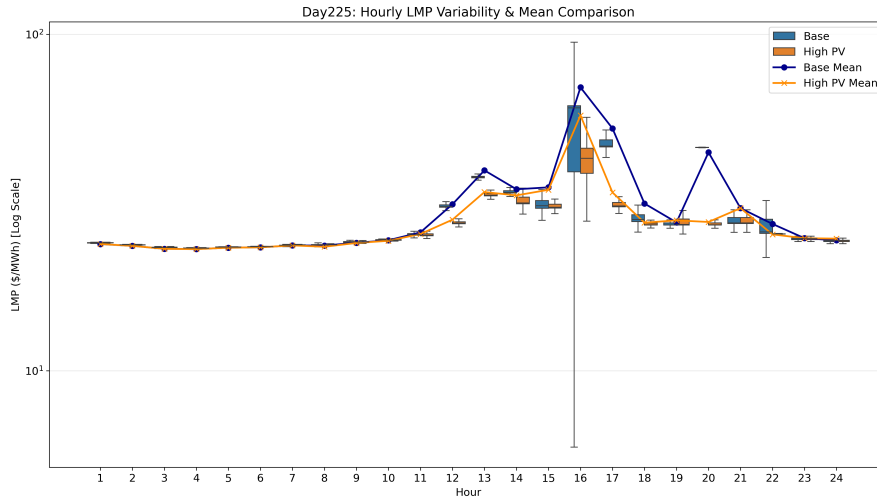


Figure 4: LMP Variability Comparison (Base vs High PV). The High PV case exhibits reduced median prices.

An examination of generator dispatch indicates that this cost penalty is driven primarily by thermal inflexibility rather than transmission congestion. Detailed post-optimization analysis reveals that the grid struggles to accommodate additional solar energy because thermal generators are constrained by their minimum generation limits (P_{min}). The number of thermal unit-hours in which generators bind at P_{min} increases by 78 hours relative to the Base Case (from 568 to 646 hours). Because these thermal units

cannot ramp down further, the system is forced to curtail approximately 2,633 MWh of available solar generation. Consequently, the cost increase reflects not an inability to move power (congestion), but an inability to shut down expensive generation, leaving the system paying for fuel it does not strictly need while wasting renewable energy.

4 Optimization Modeling of Storage Collocation

We see in previous sections that increased utility-scale solar generation creates both benefits and challenges for the grid. Although more solar power reduces LMP variability, it may be difficult to balance with incumbent thermal generators that face relatively inflexible ramping constraints. As a result, total system costs may increase even though solar generation provides low-cost power. To alleviate this concern, we use the Most Optimal Scheduling Tool (MOST) developed by Murillo-Sánchez et al. (2013) to model utility scale storage. The optimization problem is the same as before, and the storage constraints are modeled as follows:

$$p^{ti} = p_{sc}^{ti} + p_{sd}^{ti} \quad (3)$$

$$p_{sc}^{ti} \leq 0 \quad (4)$$

$$p_{sd}^{ti} \geq 0 \quad (5)$$

$$s_{-}^{ti} \geq S_{min}^{ti} \quad (6)$$

$$s_{+}^{ti} \leq S_{max}^{ti} \quad (7)$$

where p^{ti} indicates the active injection for unit i at time t , p_{sc}^{ti} indicates the amount of power being drawn by a charging storage unit i , and p_{sd}^{ti} indicates the amount of power being discharged by storage unit i . The variables s_{-}^{ti} and s_{+}^{ti} indicate the lower and upper bounds on the energy stored, respectively, in unit i at time t . Similarly, the variables S_{min}^{ti} and S_{max}^{ti} are the minimum and maximum storage units, respectively, for unit i at time t .

While high solar penetration generally reduces system costs (as demonstrated by Days 194 and 216 in Appendix B), our analysis of Day 225 highlighted a specific scenario where existing thermal generation inflexibility led to significant solar curtailment and an overall increase in system costs. To address these operational challenges and further maximize the value of renewable generation, we investigated the role of utility-scale battery storage. We tested two distinct deployment strategies: collocating storage with the new solar generation (Strategy A) versus collocating storage at high-load centers (Strategy B). Many studies have explored the benefits of Strategy A, including McLaren et al. (2019), which has contributed to the massive expansion of solar+storage projects in Texas. Recently, the power firm RWE commissioned the Stoneridge Solar PV project that adds 200 MW of solar and 100MW/200MWh of storage.³ Strategy B is a viable alternative strategy since it would alleviate transmission constraints by enabling high load buses to be served during peak hours at a sub-transmission level.

To develop Strategy A, we locate all of the solar resources in the synthetic dataset and pair them with a utility-scale battery with the capacity to deliver the same amount of power as the collocated solar generator's maximum production amount $P_{max}^{solar,i}$. We then multiply this total capacity $C_{max}^{storage,i}$ by 1.2

³RWE, <https://www.rwe.com/en/press/rwe-clean-energy/2025-11-26-rwe-brings-more-american-generated-power-to-texas-with-stoneridge-project/>, Nov. 2025

such that 10% of the battery's overall capacity can be allocated for 10-minute and 30-minute reserves as was used in the example storage case provided by Murillo-Sánchez et al. (2013).

$$P_{max}^{solar,i} = P_{max}^{storage,i} \quad (8)$$

$$C_{max}^{storage,i} = 1.2 * (4 * P_{max}^{solar,i}) \quad (9)$$

Additionally, to model charging, the minimum capacity of each battery was defined as follows:

$$C_{min}^{storage,i} = -C_{max}^{storage,i} \quad (10)$$

This ensures that the battery can be charged and discharged as determined by the optimization procedure. In addition, we assume the marginal cost of the discharge to be zero, similar to that of renewables.

We use the same procedure outlined above to develop Strategy B by selecting buses in the 90th percentile of load and then defining collocated storage for those buses. Since there are clearly more buses that fit this criteria than solar generators, we restrict the total amount of storage in Strategy B to equal the total amount of storage in Strategy A. Therefore, the amount of storage collocated on each 90th percentile load bus is proportional to the amount of load being served, but the total battery capacity does not exceed that which exists in Strategy A. Table 2 summarizes the added storage units:

Variable	Load Collocation	PV Collocation
Number of Batteries	100	36
Total Storage Capacity [MW]	11808.96	11808.96
Largest Individual Unit Capacity [MW]	185.2601	1200
Median Individual Unit Capacity [MW]	111.7818	192.72
Smallest Individual Unit Capacity [MW]	93.18506	6.72

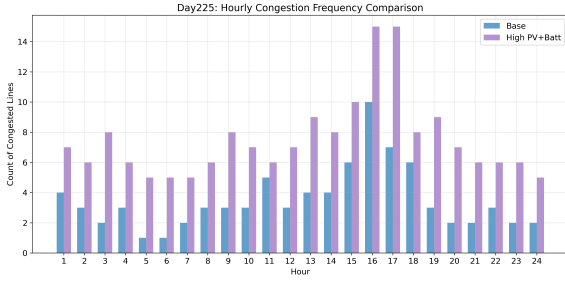
Table 2: Utility-Scale Storage Summary for Load and PV Collocation

4.1 Strategy A: Co-location with Solar Generation

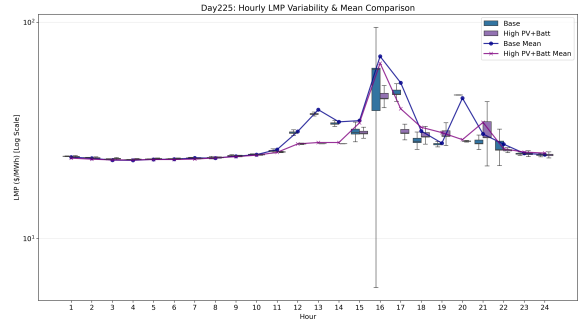
This strategy successfully reduced the Total System Cost to \$37,700,862, a savings of approximately \$97,000 (-0.26%) compared to the Base Case.

The storage units were able to absorb the curtailed excess solar energy during the day, effectively bypassing the thermal P_{min} constraints that led to the cost increase in the Day 225 High PV scenario. However, this strategy introduced a new challenge: transmission congestion. The number of congestion events more than doubled, rising from 84 in the Base Case to 180.

This result highlights a critical limitation of generation-sited storage. While the batteries could smooth the temporal intermittency of solar and reduce curtailment, discharging that stored energy from remote locations still requires utilizing the transmission network during peak hours. We hypothesize that the morning solar ramp-up caused initial congestion, and the evening battery discharge created a "second wave" of congestion, stressing the lines twice daily. This hypothesis though, not proven, aligns with observations of the Energy Systems Integration Group (2024) Group (2024), which noted similar pattern changes in transmission flows.



(a) Congestion Profile (Strategy A)



(b) LMP Variability (Strategy A)

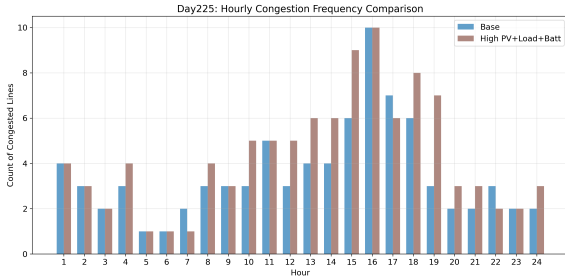
Figure 5: Co-locating batteries with solar reduced costs, reduced LMP variability, but doubled transmission congestion events.

4.2 Strategy B: Co-location with Load

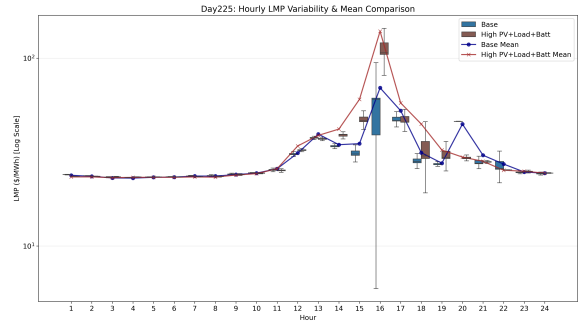
In the second strategy, we moved the identical storage capacity to high-load buses. This approach yielded the most significant benefits, reducing the Total System Cost by \$238,851 (-0.63%) compared to the Base Case.

Crucially, this strategy avoided the massive congestion penalty seen in Strategy A. Congestion events rose only slightly to 103 (compared to 180 in Strategy A). By siting batteries near the load, the system could charge using cheap off-peak power and discharge locally during peaks, relieving the strain on long-distance transmission lines. Danish et al. (2020)

Interestingly, the average LMP in this scenario *increased* to \$35.05/MWh. This counterintuitive rise in average price alongside a decrease in total cost suggests that the market signals allocated the costs more efficiently across all of the buses. The batteries likely supported prices during off-peak times (charging) and capped them during peaks, but the locational value of energy was better preserved.



(a) Congestion profile for Strategy B



(b) LMP Variability. (Strategy B)

Figure 6: Co-locating batteries with high load buses reduced LMP variability even more, even with an increased average compared to strategy A.

5 Discussion and Policy Implications

Our results highlight a nuanced reality for grid decarbonization. Our econometric findings and the simulations for typical days (Appendix C) demonstrate that increased renewable capacity, particularly solar, generally leads to reduced system costs and lower LMPs. However, the "High PV" scenario serves as a critical stress-test, illustrating that if the grid lacks sufficient flexibility, existing thermal generation

constraints can limit the full economic benefits of abundant solar and even introduce operational challenges like increased congestion. This confirms our findings that while utility-scale solar generation puts a downward pressure on cost, complementary technologies are crucial for smoothening LMP variability and producing a more cost-efficient system.

The Value of Flexibility: Storage acts as the essential "flexibility enabler." By absorbing excess generation during periods of high renewable output, it resolves the P_{min} conflict, converting curtailed energy into usable supply. This not only prevents economic losses but also enhances grid stability.

The Siting Imperative: Perhaps the most critical finding for policymakers is the sensitivity of these benefits to location.

- Generation-sited storage effectively solves the intermittency problem by smoothing the output of renewable generation. However, as demonstrated by Strategy A, it can exacerbate the transmission problem, leading to increased congestion if located far from load centers and discharged during peak hours. This approach treats the symptoms of solar variability but often ignores the critical constraints of the transmission network.
- Load-sited storage offers a more comprehensive solution. As shown in Strategy B, it acts as a "virtual transmission line," allowing the grid to bypass congestion limits by sourcing power locally during peaks. This strategy not only mitigates congestion but also improves overall system efficiency and affordability.

Policy incentives, such as the Investment Tax Credit (ITC), should therefore consider not just the *quantity* of storage deployed, but its *location*. Incentivizing storage solely at generation sites (often easier for developers due to land availability and direct connection benefits) may lead to suboptimal grid outcomes and higher transmission upgrade costs in the long run. Strategic planning that prioritizes load-sited storage can unlock greater economic and operational benefits for the grid.

6 Future Work and Conclusions

Our analysis focused on three high-load days within one year in Texas. Future work could extend this study by incorporating a full year of data to examine how seasonal variation affects the results. Additional extensions include introducing stochastic representations of renewable generation and load to capture more realistic grid conditions. One crucial component of this idea not discussed in our work is an evaluation of how battery capital costs influence the financial viability of collocating storage with high-load buses. Similarly, explicitly modeling battery degradation and performing N-1 contingency analysis would provide a more comprehensive assessment of grid reliability and long-term asset value.

Overall, this study combined econometric analysis with grid optimization to investigate the relationship between renewables, storage, and electricity affordability. Our econometric results challenge the narrative that renewables drive up prices, showing a correlation between utility-scale solar and lower rates. Our optimization modeling generally confirms our initial findings that during sub-peak load conditions, additional solar generation may lower total grid costs. However, this also may produce high system-wide LMP variability, which may cause utilities to set retail rates higher than if LMPs were lower and less volatile. While solar alone faces integration challenges (the inflexibility penalty), strategic storage deployment unlocks its value. Collocating storage with high-load buses alleviates grid constraints and results in less volatile LMPs with along with lower overall system costs. We conclude that the pathway to an affordable, renewable-heavy grid lies in the strategic deployment of storage, specifically targeted near load centers to maximize transmission efficiency.

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Author Contributions

Conceptualization, F.A. and I.T.; econometric analysis, F.A.; MatPower and MOST optimization, F.A.; report writing, F.A. and I.T.; optimization results analysis, I.T.; slides, F.A. and I.T.

A Appendix: Econometric Analysis

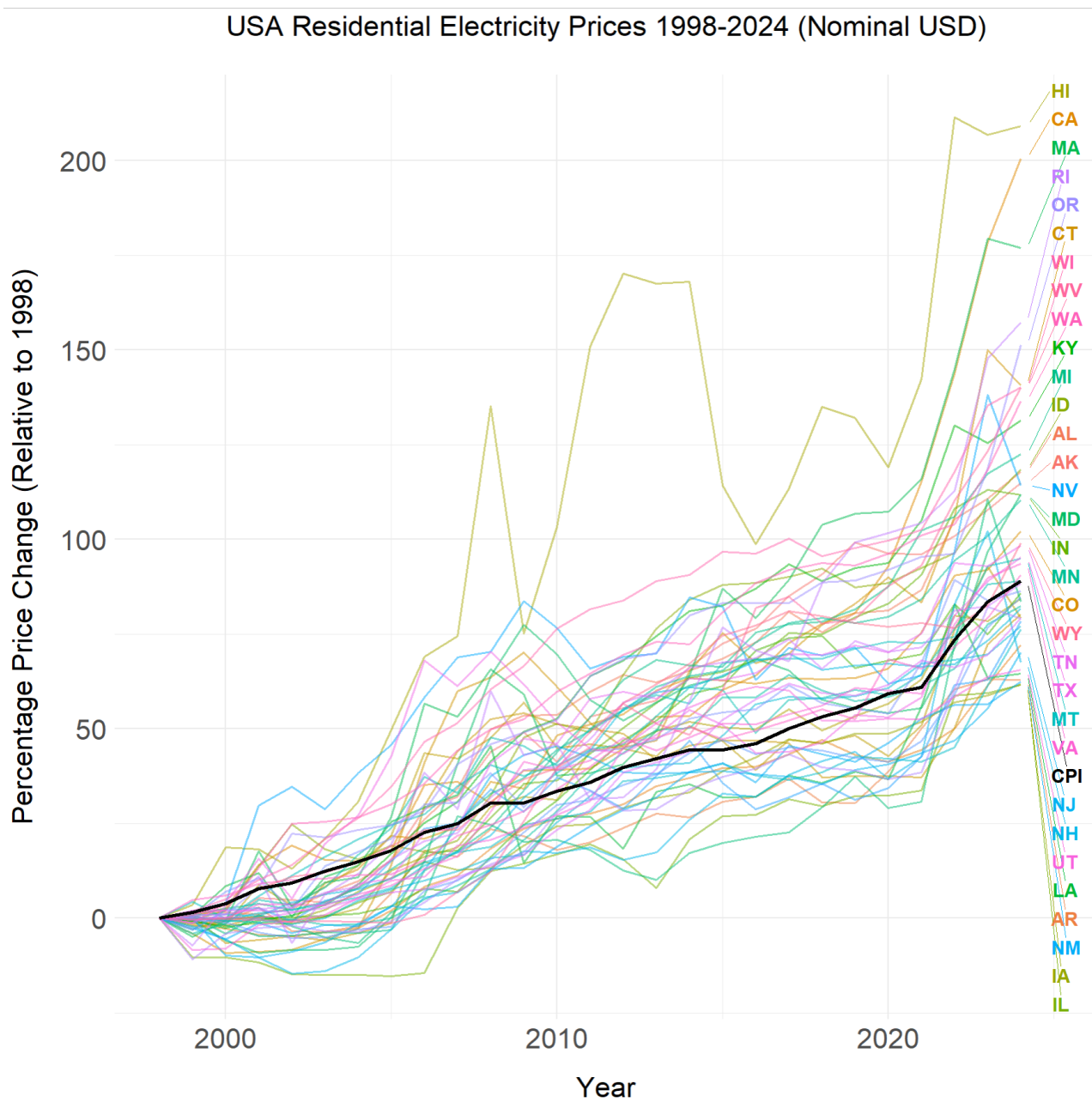


Figure A1: This plot visualizes the growth of electricity rates in the United States from 1998 through 2024. The data comes from the EIA. Half of the states experience prices above the CPI.

Table A1: Variable definitions, units, usage, and data sources for the econometric analysis. All data is publicly available and published by the Federal Energy Regulatory Commission (FERC) and the Energy Information Administration (EIA)

Variable	Units	Usage	Source
Residential electricity price	Revenue in cents/kWh sold to residential consumers	Outcome variable	EIA Form 860
O&M generation, transmission, and distribution costs	Revenue in cents/total kWh of electricity sold	Outcome variable	FERC Form 1
CapEx generation, transmission, and distribution	Revenue in cents/total kWh of electricity sold	Outcome variable	FERC Form 1
Utility PV & thermal generation	Share of total generation	Variable of interest	EIA Form 923
Residential PV generation	Share of total generation	Variable of interest	EIA Form 923
Wind generation	Share of total generation	Variable of interest	EIA Form 923
Hydro generation	Share of total generation	Variable of interest	EIA Form 923
Pumped storage generation	Share of total generation	Control	EIA Form 923
Natural gas generation	Share of total generation	Control	EIA Form 923
Nuclear generation	Share of total generation	Control	EIA Form 923
Petroleum generation	Share of total generation	Control	EIA Form 923
Other generation	Share of total generation	Control	EIA Form 923

Table A2: Impact of RPS and Source Generation Shares on CONUS Residential Electricity Price (1998-2023), in nominal \$

	(1)	(2)	(3)	(4)
Utility PV & Thermal Generation	−1.1552+ (0.6495)	−1.9919** (0.6711)	0.3812 (0.7982)	−0.8010+ (0.4132)
Residential PV Generation	13.3615*** (1.9436)	11.7423*** (1.9151)	7.0936** (2.4863)	2.8530+ (1.6543)
Wind Generation	1.1746*** (0.1390)	0.4364** (0.1277)	1.2946*** (0.1752)	−0.1208 (0.1062)
Hydro Generation	−0.1362 (0.0963)	−0.1414 (0.0942)	−0.0343 (0.3021)	−0.2131* (0.0945)
Pumped Storage Generation	2.0497 (5.2017)	1.7179 (5.4073)	4.5516 (4.9148)	−5.5102+ (3.1397)
Natural Gas Generation	0.5946*** (0.0853)	0.3725** (0.1081)	0.8695*** (0.1435)	−0.1318 (0.0864)
Nuclear Generation	0.6633*** (0.0993)	0.5452*** (0.0975)	0.6123** (0.1942)	0.0727 (0.0918)
Petroleum Generation	−0.1073 (0.3244)	0.9147* (0.3604)	−0.3323 (0.2505)	−0.0993 (0.2688)
Other Generation	1.6090*** (0.2786)	1.6496*** (0.2797)	1.2936+ (0.7051)	−0.0410 (0.4633)
R2 Adj.	0.602	0.755	0.795	0.946
Num.Obs.	1248	1248	1248	1248
R2	0.605	0.762	0.804	0.950
Year FE		Yes		Yes
State FE			Yes	Yes

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Dependent Variable: Log residential electricity price [¢/kwh], in nominal USD. State clustered standard errors. The omitted comparison group is the share of coal generation. Generation is expressed in MWh as a share of the total generation for each state in a particular year.

Table A3: Impact of Source Generation Shares on Distribution Costs (1998-2023), in nominal \$

	(1)	(2)	(3)	(4)	(5)
Residential PV	26.3263** (8.4795)	23.1681* (9.3540)	14.8509* (6.5365)	10.8634*** (2.5374)	6.3430* (2.5718)
Utility PV & Thermal Generation	−6.1496 (3.9714)	−7.1481+ (4.2150)	0.1085 (2.2653)	0.5860 (0.9169)	−2.3832* (1.0349)
Wind Generation	1.5856*** (0.3542)	0.4014 (0.4853)	1.3029* (0.5044)	1.2394*** (0.3326)	−0.8007* (0.3047)
Hydro Generation	0.8542** (0.2956)	0.8243** (0.2854)	0.5864 (0.9485)	0.0641 (0.1076)	0.0195 (0.3649)
Pumped Storage Generation	0.0890 (10.5017)	−1.3321 (10.5425)	−14.5478+ (7.2697)	6.9736** (2.3371)	−11.2608 (8.7503)
Natural Gas Generation	0.8738** (0.2842)	0.4464 (0.2778)	1.2375*** (0.3099)	0.7993*** (0.1630)	−0.2779 (0.2743)
Nuclear Generation	1.0419* (0.4715)	0.8709+ (0.4867)	1.2440+ (0.6878)	0.2332* (0.1042)	−0.0687 (0.4390)
Petroleum Generation	−0.6862 (1.0029)	0.6332 (1.1637)	−1.5771* (0.7059)	−1.8015** (0.5936)	−0.6073 (0.7857)
Other Generation	3.5866*** (0.6987)	3.8816*** (0.7327)	1.1371 (2.9914)	−0.6264 (0.4749)	−0.8761 (2.0719)
R2 Adj.	0.078	0.093	0.177	0.874	0.895
Num.Obs.	7417	7417	7417	7417	7417
R2	0.079	0.097	0.183	0.877	0.899
Year FE		Yes			Yes
State FE			Yes		Yes
Utility FE				Yes	Yes

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Dependent Variable: Logged O&M distribution costs per-kWh total electricity sales, in nominal USD [¢/kWh]. State clustered standard errors.

Table A4: Impact of Source Generation Shares on Transmission Costs (1998-2023), in nominal \$

	(1)	(2)	(3)	(4)	(5)
Residential PV	29.5085*** (8.2362)	22.9625* (9.5904)	17.3700* (7.1109)	11.8087** (3.9436)	5.1600 (5.3706)
Utility PV & Thermal Generation	-9.5133** (3.4382)	-11.9053** (3.8051)	-2.4807 (2.4386)	-0.4875 (1.5279)	-4.9396* (2.1534)
Wind Generation	4.7882*** (0.4847)	2.2568*** (0.3928)	4.2102*** (0.5470)	3.3850*** (0.4133)	0.7397 (0.5083)
Hydro Generation	0.5367+ (0.3011)	0.4779+ (0.2573)	1.5883* (0.7633)	0.0932 (0.1984)	0.8727* (0.3988)
Pumped Storage Generation	-19.0179 (24.3253)	-20.0758 (24.5981)	1.7680 (15.0092)	12.5348* (5.0636)	-16.6471 (18.7938)
Natural Gas Generation	0.8732* (0.3461)	-0.0124 (0.3456)	2.9066*** (0.3583)	1.5349*** (0.2776)	-0.0210 (0.5610)
Nuclear Generation	0.1205 (0.4684)	-0.2478 (0.4839)	2.5961*** (0.6817)	0.6678** (0.1940)	0.7058 (0.5574)
Petroleum Generation	-2.4761 (1.6505)	0.8915 (1.7151)	-1.0007 (1.2345)	-2.7944*** (0.7053)	0.0301 (1.3388)
Other Generation	6.1199** (2.0212)	6.6883*** (1.4685)	6.5948*** (1.8303)	-0.3172 (0.8890)	0.5363 (1.2945)
R2 Adj.	0.194	0.290	0.372	0.741	0.824
Num.Obs.	7579	7579	7579	7579	7579
R2	0.195	0.293	0.376	0.747	0.830
Year FE		Yes			Yes
State FE			Yes		Yes
Utility FE				Yes	Yes

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Dependent Variable: Logged O&M transmission costs per-kWh total electricity sales, in nominal USD [¢/kWh]. State clustered standard errors.

Table A5: Impact of Source Generation Shares on Generation Costs (1998-2023), in nominal \$

	(1)	(2)	(3)	(4)	(5)
Residential PV	-16.5059 (13.6690)	-16.8738 (13.5724)	9.9126 (8.0879)	-2.6801 (4.4300)	-1.2008 (5.6015)
Utility PV & Thermal Generation	4.8336 (4.8882)	4.8799 (4.9898)	-5.3403* (2.2678)	1.2190 (1.2052)	-0.4520 (1.4961)
Wind Generation	0.1914 (0.3162)	-0.2779 (0.4529)	0.2108 (0.2896)	0.3718+ (0.1939)	-0.5897 (0.4200)
Hydro Generation	-0.6347 (0.4025)	-0.6158 (0.4134)	-2.2351** (0.6402)	-0.1732** (0.0636)	-2.5523*** (0.6936)
Pumped Storage Generation	12.2219 (17.8004)	10.5042 (17.6710)	20.4419 (14.9135)	7.6154+ (4.2045)	-0.5382 (15.9950)
Natural Gas Generation	0.6650 (0.3962)	0.5138 (0.4112)	1.1633*** (0.3136)	0.2149+ (0.1198)	0.2340 (0.5352)
Nuclear Generation	-0.9976* (0.4949)	-1.0252* (0.4903)	0.6972 (0.9509)	0.1387 (0.1400)	-0.5749 (0.7766)
Petroleum Generation	-2.8830 (2.4376)	-1.8755 (2.5226)	1.9551+ (1.1064)	2.6525* (1.2152)	4.2972** (1.4112)
Other Generation	-6.8322** (2.3569)	-7.0249** (2.4305)	7.3809* (3.3452)	-0.2712 (0.3627)	0.9508 (2.3851)
R2 Adj.	0.075	0.092	0.219	0.574	0.591
Num.Obs.	6740	6740	6740	6740	6740
R2	0.077	0.096	0.226	0.585	0.606
Year FE		Yes			Yes
State FE			Yes		Yes
Utility FE				Yes	Yes

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Dependent Variable: Logged O&M generation costs per-kWh total electricity sales, in nominal USD [¢/kWh]. State clustered standard errors.

Table A6: Impact of Source Generation Shares on Total Generation and Delivery Costs (1998-2023),
in nominal \$

	(1)	(2)	(3)	(4)	(5)
Residential PV	1.0340 (5.4320)	-0.8932 (5.4675)	10.1483*** (2.4042)	9.5224*** (2.2304)	6.7700* (2.9873)
Utility PV & Thermal Generation	1.4056 (2.0577)	0.9163 (2.0002)	-1.2424 (1.0788)	-0.5130 (0.9510)	-3.0236** (1.0874)
Wind Generation	1.3891*** (0.2347)	0.5379* (0.2525)	1.2235*** (0.3288)	1.1426*** (0.2264)	-0.4822* (0.2209)
Hydro Generation	-0.3365* (0.1307)	-0.3365* (0.1293)	-0.5210 (0.6944)	-0.0720 (0.0704)	-0.8468* (0.3892)
Pumped Storage Generation	2.9138 (7.9611)	2.2334 (8.1071)	21.3607** (6.7827)	6.5674** (1.9188)	2.4592 (8.4800)
Natural Gas Generation	0.4897** (0.1402)	0.1893 (0.1526)	0.9430*** (0.1968)	0.5232*** (0.0908)	-0.5656* (0.2594)
Nuclear Generation	-0.5839* (0.2704)	-0.6985* (0.2701)	0.0632 (0.3916)	0.1720* (0.0689)	-0.2830 (0.3303)
Petroleum Generation	-1.8801+ (0.9426)	-0.4587 (0.9625)	-1.3360+ (0.7376)	-1.9676*** (0.5030)	-1.2034 (0.8474)
Other Generation	1.3381* (0.5207)	1.3665** (0.4935)	1.4555 (1.3087)	-0.1111 (0.3375)	0.2116 (1.6790)
R2 Adj.	0.094	0.138	0.218	0.692	0.740
Num.Obs.	7868	7868	7868	7868	7868
R2	0.095	0.141	0.224	0.700	0.749
Year FE		Yes			Yes
State FE			Yes		Yes
Utility FE				Yes	Yes

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

Dependent Variable: Logged O&M total generation and delivery costs per-kWh total electricity sales,
in nominal USD [¢/kWh]. State clustered standard errors.

B Appendix: Optimization Formulation, adapted from Lee et al. (2022)

Table A7: Decision variables in SCUC and SCED

Variable	Description
$g_i[t]$	MW output of generator i at time t
$p_i[t]$	Net MW power injection at bus i at time t
$\theta_i[t]$	Voltage angle of bus i at time t
$f_{ij}[t]$	Power flow on line (i, j) at time t

Table A8: Decision variables in SCUC

Variable	Description
$z_i[t] \in \{0, 1\}$	Commitment decision; $z_i[t] = 1$ if generator i is on at time t
$u_i[t] \in \{0, 1\}$	Startup indicator; $u_i[t] = 1$ if generator i is turned on at time t
$v_i[t] \in \{0, 1\}$	Shutdown indicator; $v_i[t] = 1$ if generator i is turned off at time t

Table A9: Parameters and constants

Symbol	Description
$c_i^g(\cdot)$	Piecewise linear or quadratic cost function of generator i
c_i^z	No-load cost of generator i
c_i^u	Startup cost coefficient of generator i
$d_i[t]$	Forecast of load at bus i at time t
$\bar{g}_i, g_i$	MW upper and lower bounds of generator i
$g_i[0]$	Initial generation level of generator i at $t = 0$
$\bar{\theta}_i, \underline{\theta}_i$	Upper and lower bounds on voltage angle at bus i
$\bar{r}_i, r_i$	Ramping upper and lower bounds of generator i
$\bar{f}_{ij}, \underline{f}_{ij}$	Upper and lower bounds on flow of line (i, j)
$\underline{u}_i, \underline{v}_i$	Minimum on and off time for generator i
x_{ij}	Reactance of line (i, j)

B.1 Security-Constrained Unit Commitment (SCUC)

$$\text{minimize } \sum_{t=1}^T \sum_{i \in \mathcal{G}} c_i^g (g_i[t]) + c_i^z z[t] + c_i^u u[t] + c_i^v v[t] \quad (\text{Equation2a})$$

$$\text{subject to } p_i[t] = g_i[t] - d_i[t] \quad : (\lambda_i[t]) \quad i \in \mathcal{N}; t = 1, 2, \dots, T. \quad (\text{Equation2b})$$

$$p_i[t] = \sum_{(i,j) \in \mathcal{E}} f_{ij}[t] \quad i \in \mathcal{N}; (i,j) \in \mathcal{E}; t = 1, 2, \dots, T. \quad (\text{Equation2c})$$

$$\underline{f}_{ij} \leq f_{ij}[t] \leq \bar{f}_{ij} \quad : (\mu_{ij}[t]) \quad (i,j) \in \mathcal{E}; t = 1, 2, \dots, T. \quad (\text{Equation2d})$$

$$f_{ij}[t] = \frac{\theta_i[t] - \theta_j[t]}{x_{ij}} \quad (i,j) \in \mathcal{E}; t = 1, 2, \dots, T. \quad (\text{Equation2e})$$

$$\underline{r}_i \leq g_i[t] - g_i[t-1] \leq \bar{r}_i \quad i \in \mathcal{G}; t = 1, 2, \dots, T. \quad (\text{Equation2f})$$

$$z_i[t] \cdot \underline{g}_i \leq g_i[t] \leq z_i[t] \cdot \bar{g}_i \quad i \in \mathcal{G}; t = 1, 2, \dots, T. \quad (\text{Equation2g})$$

$$u_i[t] \geq z_i[t] - z_i[t-1] \quad i \in \mathcal{G}; t = 1, 2, \dots, T. \quad (\text{Equation2h})$$

$$v_i[t] \geq z_i[t-1] - z_i[t] \quad i \in \mathcal{G}; t = 1, 2, \dots, T. \quad (\text{Equation2i})$$

$$z_i[t] - z_i[t-1] \leq z_i[\iota], \quad \iota \in [t+1, \min\{t + \underline{u}_i - 1, T\}], \quad t = 2, 3, \dots, T. \quad (\text{Equation2j})$$

$$z_i[t-1] - z_i[t] \leq 1 - z_i[\iota], \quad \iota \in [t+1, \min\{t + \underline{v}_i - 1, T\}], \quad t = 2, 3, \dots, T. \quad (\text{Equation2k})$$

Figure A2: SCUC optimization problem formulation as defined by Lee et al. (2022). This is the code we iterate from for our project.

B.2 Security-Constrained Economic Dispatch (SCED)

$$\text{minimize} \quad \sum_{t=1}^{n_t} \sum_{i \in \mathcal{G}} c_i^g (g_i [t]) \quad (\text{Equation 1a})$$

$$\text{subject to} \quad p_i [t] = g_i [t] - d_i [t] : \quad (\lambda_i [t]) \quad i \in \mathcal{N}; t = 1, 2, \dots, n_t. \quad (\text{Equation 1b})$$

$$f_{ij} [t] = \frac{\theta_i [t] - \theta_j [t]}{x_{ij}} \quad (i, j) \in \mathcal{E}; t = 1, 2, \dots, n_t. \quad (\text{Equation 1c})$$

$$p_i [t] = \sum_{(i,j) \in \mathcal{E}} f_{ij} [t] \quad i \in \mathcal{N}; (i, j) \in \mathcal{E}; t = 1, 2, \dots, n_t. \quad (\text{Equation 1d})$$

$$\underline{f}_{ij} \leq f_{ij} [t] \leq \bar{f}_{ij} \quad : (\mu_{ij} [t]) \quad (i, j) \in \mathcal{E}; t = 1, 2, \dots, n_t. \quad (\text{Equation 1e})$$

$$\underline{\theta}_i \leq \theta_i [t] \leq \bar{\theta}_i \quad i \in \mathcal{N}; t = 1, 2, \dots, n_t. \quad (\text{Equation 1f})$$

$$\underline{r}_i \leq g_i [t] - g_i [t-1] \leq \bar{r}_i \quad i \in \mathcal{G}; t = 1, 2, \dots, n_t. \quad (\text{Equation 1g})$$

$$\underline{g}_i \leq g_i [t] \leq \bar{g}_i \quad i \in \mathcal{G}; t = 1, 2, \dots, n_t. \quad (\text{Equation 1h})$$

Figure A3: SCED optimization problem formulation as defined by Lee et al. (2022). This is the code we iterate from for our project.

C Appendix: Extended High PV Analysis

C.1 Day 194 High PV Comparison

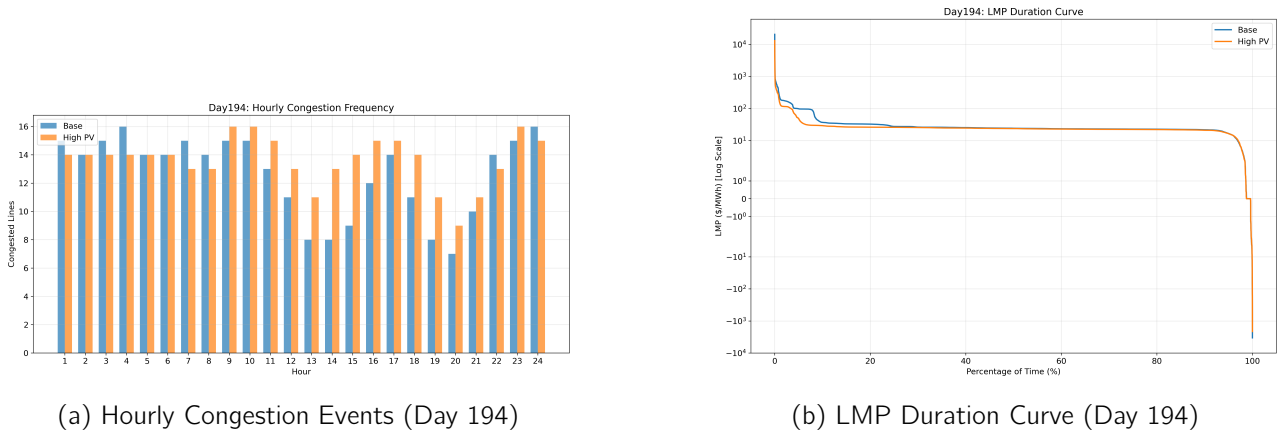


Figure A4: High PV Case operational metrics showing reduced congestion and price stability.

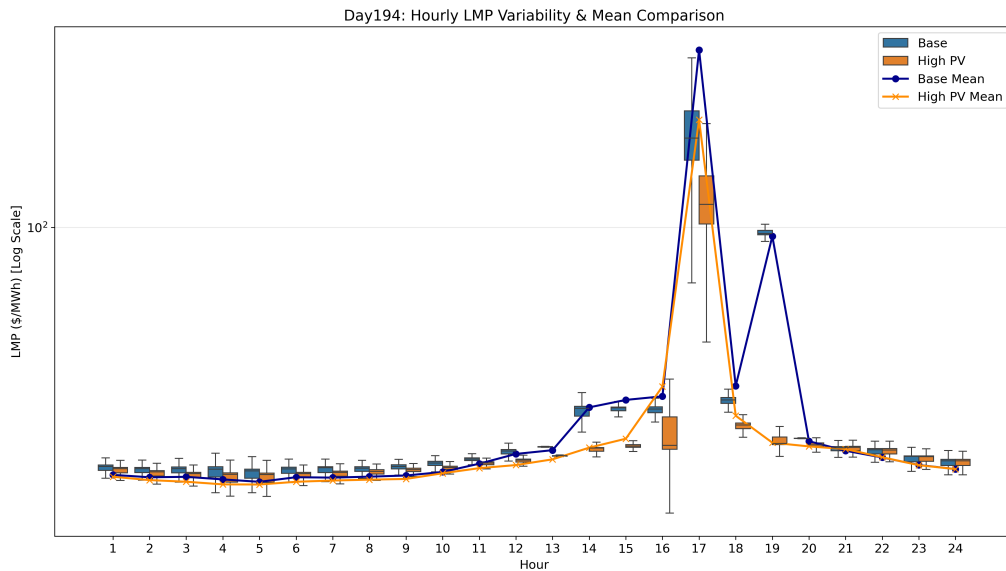


Figure A5: LMP Variability Comparison. The High PV case exhibits reduced median prices and less variability.

C.2 Day 216 High PV Comparison

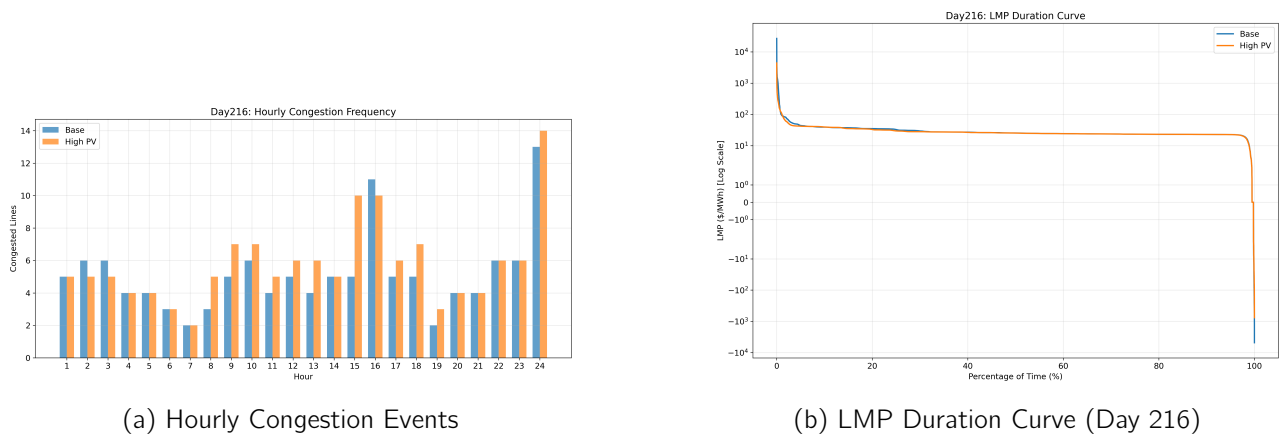


Figure A6: High PV Case operational metrics showing reduced congestion and price stability.

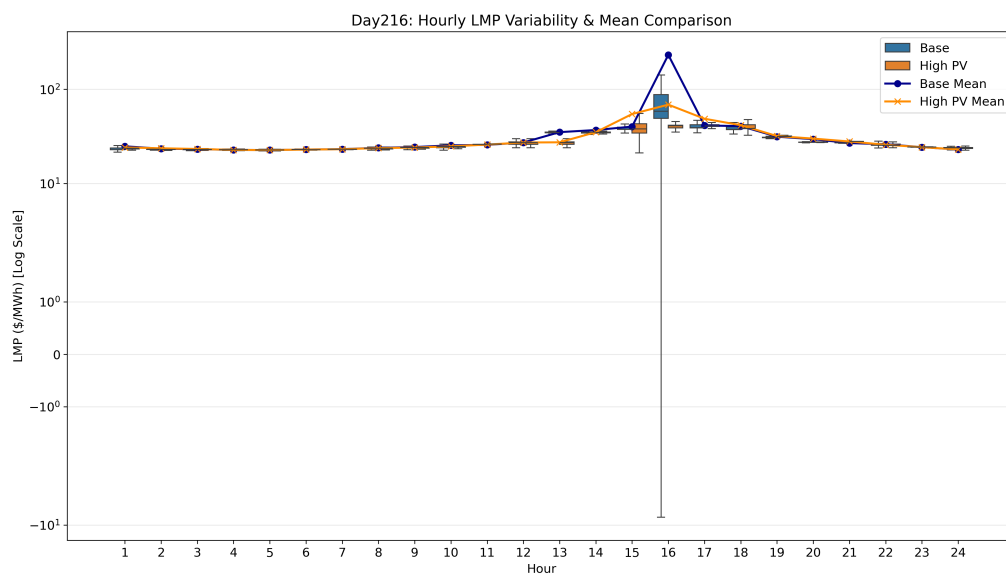


Figure A7: LMP Variability Comparison. The High PV case exhibits reduced median prices and less variability.